

BOOSTING MARITIME CYBER RESILIENCE THROUGH DEEP LEARNING

Review-6 Report

**Bachelor of Technology in
Computer Science and Engineering
(Cyber Security)**

by

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Computer Science and Engineering (Cyber Security)

**INSTITUTE OF AERONAUTICAL ENGINEERING
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BOOSTING MARITIME CYBER RESILIENCE THROUGH DEEP LEARNING

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DECLARATION

I certify that

- a. The work contained in this report is original and has been done by me under the guidance of my supervisor(s).
- b. The work has been submitted to any other Institute for any degree or diploma.
- c. I have followed the guidelines provided by the Institute in preparing the report.
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Signature of the Students

Date:

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CERTIFICATE

This is to certify that the project report entitled **Boosting Maritime Cyber Resilience through deep learning** submitted by **B. Sameera, D. Geethamsh Rao and P. Raj Kumar** to the Institute of Aeronautical Engineering, Hyderabad in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in **CSE(Cyber Security)** is a bonafide record of work carried out by him/her under my/our guidance and supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute for the award of any Degree.

Supervisor
(Mr. B Santhosh Kumar)

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Date:

APPROVAL SHEET

This project report entitled **Boosting Maritime Cyber Resilience through Deep Learning** by **B. Sameera, D. Geethamsh Rao and P. Raj Kumar** is approved for the award of the Degree Bachelor of Technology in **CSE (Cyber Security)**.

Examiners

Supervisor(s)

Mr. B Santhosh Kumar

Principal

Dr. L V Narasimha Prasad

Date:

Place:

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ABSTRACT

The integration of Internet of Things (IoT) technologies into Maritime Transportation Systems (MTS) has brought about significant changes, but it has also led to increased cybersecurity risks due to the expanded connectivity. To address these challenges, the proposed Deep Learning Threat Intelligence Framework (DLTIF) is designed to enhance the security of IoT-enabled MTS. DLTIF consists of three main components: the Deep Feature Extractor (DFE), Cyber Threat Intelligence-Driven Detection (CTIDD), and Cyber Threat Intelligence Attack Type Identification (CTIATI). Utilizing Convolutional Neural Networks (CNN) alongside ensemble techniques like CNN+LSTM and Stacking Classifier, the framework delivers impressive accuracy in identifying threats, outperforming conventional methods and reinforcing MTS security. The Stacking Classifier, in particular, shows improved accuracy, highlighting the strength of ensemble techniques. Additionally, a Flask-based interface simplifies the testing process, with built-in authentication mechanisms ensuring only authorized access, thus safeguarding the system and maintaining data integrity. By combining advanced machine learning models with a user-friendly interface, DLTIF strengthens the protection of critical infrastructure, enhancing both threat detection capabilities and resilience against emerging cyber risks.

Keywords: Deep Learning Threat Intelligence Framework (DLTIF), IoT, Maritime Transportation Systems (MTS), CTI-Attack Type Identification (CTIATI), CNN.